

floorplan.tcl

Description : Floorplanning script for a placed-and-routed design.

Tool : Synopsys ICC2 (or compatible implementation tool)

Requirements : tech_setup.tcl, user_setup.tcl, cut_site_rows.tcl, pg_routing.tcl

Variables :

max_core - Maximum number of CPU cores to use

ndm_lib_name - Name of the MDM/NDM library in db/

design - Top-level design name

push_down_fp - Push-down floorplan file (DEF/TCL)

macro_place - Macro placement file (DEF/TCL)

stop_after_floorplan - Boolean flag; if 1, stop after floorplan

Read Technology and User Settings

```
source -e -v ./scripts/tech_setup.tcl
```

```
source -e -v ./scripts/user_setup.tcl
```

Multi-CPU Setting

```
set_host_options -max_cores $max_core
```

Open MDM/NDM Library

```
if { [file exists "db/${ndm_lib_name}"] } {  
    puts "\n[INFO] Library 'db/${ndm_lib_name}' exists.
```

```
Opening...\n"
```

```
    open_lib db/${ndm_lib_name}
```

```
    copy_block -from $(design)_import_done -to
```

```
$(design)_floorplan_done
```

```
    current_block $(design)_floorplan_done
```

```
    link
```

```
} else {
```

```
    puts "\n[ERROR] MDM Library 'db/${ndm_lib_name}'  
does not exist. Please check.\n"  
    return  
}
```

```
# Read Push-Down Floorplan (DEF/TCL)
```

```
if { [file exists $push_down_fp] && [regexp "def"  
$push_down_fp] } {  
    puts "\n[INFO] Reading push-down floorplan DEF:  
$push_down_fp\n"  
    read_def $push_down_fp  
} elseif { [file exists $push_down_fp] && [regexp "tcl"  
$push_down_fp] } {  
    puts "\n[INFO] Sourcing push-down floorplan TCL:  
$push_down_fp\n"  
    source $push_down_fp  
} else {  
    puts "\n[ERROR] Push-down floorplan file  
'$push_down_fp' does not exist. Please check.\n"  
    return  
}
```

```
# Read Macro Placement (DEF/TCL)
```

```
if { [file exists $macro_place] && [regexp "def"  
$macro_place] } {  
    puts "\n[INFO] Reading macro placement DEF:  
$macro_place\n"  
    read_def $macro_place
```

```
} elseif { [file exists $macro_place] && [regexp "tcl"
$macro_place] } {
    puts "\n[INFO] Sourcing macro placement TCL:
$macro_place\n"
    source $macro_place
} else {
    puts "\n[WARNING] Macro placement file is not
provided. Please check.\n"
    return
}
```

```
# Save Block After Macro Placement
save_block -as $(design)_macros_placed
```

```
# Cut Site Rows Around Macros
source ./scripts/cut_site_rows.tcl
```

```
# Perform Power/Ground (PG) Routing
source ./scripts/pg_routing.tcl
```

```
# Save the Block
save_block
```

```
# Mark Floorplan Completion and Exit/Continue
sh touch floorplan_done
```

```
if { $stop_after_floorplan } {
```

```
    puts "\n[INFO] Floorplan stage completed. Stopping as  
requested.\n"  
    return  
} else {  
    puts "\n[INFO] Floorplan stage completed. Proceeding  
to next stage.\n"  
    close_blocks -force  
    close_lib  
    exit  
}
```